

PRACHI SINHA

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Education

Kalinga Institute of Industrial Technology Deemed to be University

2024 – 2028 (Expected)

Bachelor of Technology in Computer Science (AI Specialization)

Bhubaneswar, Odisha

- CGPA: 8.69/10

Relevant Coursework

- Data Structures and Algorithms
- Object-Oriented Programming
- Operating Systems
- Database Management Systems
- Automata and Formal Languages
- Artificial Intelligence
- Discrete Mathematics
- Computer Architecture and Organization

Projects

Song Popularity Predictor | *Python, Scikit-learn, Flask, Render*

2026

- Developed and deployed an end-to-end ML web application to predict song popularity from audio features.
- Trained and optimized a Random Forest Regressor ($R^2 \approx 0.38$, MAE ≈ 13.7) on a dataset of 100K+ tracks.
- Implemented feature preprocessing pipeline using StandardScaler and model serialization with joblib.
- Developed interactive frontend using Flask and deployed live on Render.

Spotify Recommendation System | *Python, Pandas, Scikit-learn*

2026

- Developed a content-based recommendation system using audio feature similarity.
- Implemented cosine similarity / distance-based ranking to suggest similar songs.
- Performed feature scaling and vector-based similarity comparison across tracks.

NLP Mini Projects (Poetic and Translation Chatbots) | *Python, HuggingFace, LLM APIs*

2025

- Built a prompt-engineered poetic chatbot using a generative AI API to produce themed poetry.
- Implemented a multilingual translation chatbot using pretrained transformer models from HuggingFace.
- Designed interactive inference pipelines for text generation and translation tasks.

Spotify Data Analysis | *Python, Pandas, Matplotlib, Seaborn*

2026

- Conducted exploratory data analysis on Spotify dataset to identify patterns between audio features and popularity.
- Visualised correlations between features such as energy, valence, and danceability.
- Performed statistical analysis to derive insights into factors influencing song success.

Technical Skills

Languages: Python, Java, C, SQL
ML Libraries: Scikit-learn, Pandas, NumPy
NLP: HuggingFace Transformers, LLM APIs
Data Visualization: Matplotlib, Seaborn
Web & Deployment: Flask, Render
Tools: Git, VS Code

Certifications

Generative AI Program- Finlatics (2025)
AWS Cloud Practitioner Essentials- Amazon Web Services (2025)

Extracurricular

Ethical Hacking Workshop - IIT Bombay Techfest

December 2025

- Explored fundamentals of penetration testing and network security.
- Gained hands-on exposure to vulnerability assessment techniques.